The gearbox will remain attached to the motor. The gearbox output shaft will connect to the drum shaft through a coupling.

I recommend considering an emergency brake on the gearbox output side or directly on the drum shaft. The brake location and required torque should be determined from the gearbox capacity, output speed, and drum inertia. Do not assume the gearbox will hold the drum when power is off.

Proposed arrangement

DC geared motor → gearbox → emergency brake (proposed drum-side location) → coupling → drum shaft → cable drum

This is a conceptual arrangement. The final brake location and torque rating need to be checked against the mechanical design.

Brake on the right side of the drum

Yes, installing an emergency brake on the right side of the drum could be a good option. It may be practical to mount it directly on the drum shaft if there is enough shaft extension, mounting space, and structural support.

A possible arrangement:

- Extend the shaft from the right side of the drum.
- Fit a brake disc (rotor) to the shaft.
- Mount a brake caliper on both sides of the disc.
- Secure the caliper bracket firmly to the frame.
- When the emergency stop is activated, the brake clamps the disc and slows the drum.

Check that the disc is secured to the shaft with a suitable key or approved locking method, the caliper bracket is strong enough, and the brake does not create excessive bending load on the shaft. The brake’s stopping torque and heat capacity also need to be calculated, and there must be enough clearance around the drum.

Spring-applied, electrically released electromagnetic brake

This brake is held on by spring force in its normal state and releases when powered:

Power off = brake on; power on = brake off.

It typically contains an electromagnetic coil, a spring, an armature plate, friction surfaces, and a hub that connects the brake to the shaft.

- During normal operation: The coil is powered, creating a magnetic field. The armature plate moves against the spring force, releasing the friction surfaces so the shaft and drum can rotate.

- During an emergency or power loss: Power to the coil is cut. The magnetic field disappears, the spring presses the friction surfaces together, and braking torque acts on the shaft.

You could mount this brake on the drum's right side if there is enough shaft extension and support. However, the brake will not necessarily stop the drum instantly: the drum's rotational inertia affects stopping time. Before choosing a brake, calculate the required braking torque, maximum drum RPM, drum and cable inertia, stopping time, brake heat capacity, and holding torque against cable tension. Test and validate the brake under the expected cable tension before operating the system.

Using an ESP32 instead of an Arduino

Yes, you can use an ESP32 instead of an Arduino UNO. It can read the encoder and load cell, monitor limit switches, send speed and direction commands to the motor driver, and control the brake through a properly rated switching circuit.

Keep these points in mind:

- ESP32 GPIO pins use 3.3 V logic. Do not connect 5 V signals directly to them.

- Do not power the motor or brake coil directly from ESP32 GPIO pins.

- Use a suitable motor driver and a rated brake switching circuit.

- Keep the emergency stop's fail-safe function in a separate hardware safety circuit. It should not depend only on the ESP32's software.

A possible arrangement is:

ESP32 → motor driver → geared motor

Encoder → ESP32

Load cell → HX711 → ESP32

E-stop → hardware safety circuit → motor drive disable and brake control

Why use a load cell and HX711?

The load cell measures the force acting on the cable. The HX711 amplifies the load cell's very small electrical signal and converts it to digital data that the ESP32 can read.

Cable tension → load cell signal → HX711 → ESP32 → motor adjustment, stop, or alarm

The load cell only measures cable tension accurately when it is installed in a suitable mechanical arrangement. One practical option is to run the cable through guide rollers and mount one roller on a load-cell-supported mechanism. The ESP32 can then respond if the measured tension exceeds a calibrated limit.

A load cell measures force; the HX711 conditions and converts the signal; the ESP32 reads the data and decides what to do. Tension monitoring can help detect a snag or sudden pull, but it is not a replacement for the emergency brake.

BTS7960 or Cytron motor driver?

Choose the motor driver based on the motor’s actual current requirements, including its stall current—not just an advertised current rating.

Driver Possible use

BTS7960 (IBT-2 module) Low-cost prototype testing. Its actual continuous capacity depends on the board, cooling, and wiring.

Cytron MD20A Candidate if the motor’s required current stays within its 20 A continuous rating.

Cytron MD30C Candidate if the motor’s required current stays within its 30 A continuous rating.

The earlier suggestion was to consider the Cytron MD30C, but it should not be treated as the final choice until the motor’s rated and stall current are confirmed. An industrial DC motor controller with current limiting and fault feedback may also be worth considering. The motor driver and mechanical emergency brake serve different purposes.

To choose the driver, check the motor’s nameplate or datasheet for its rated current and stall current. If those figures aren’t available, the motor’s model number or a photo of its nameplate would help.